

IPMH: Integration of stepped care for perinatal mood and anxiety disorders among women attending MCH clinics

Data and Safety Monitoring Board (DSMB): Open Report

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1. Study summary

Title	IPMH: Integration of stepped care for perinatal mood and anxiety disorders among women attending MCH clinics
Primary Objective	Integrate IPMH, a stepped care intervention for screening and treatment of perinatal mood and anxiety disorders (PMADs), and its implementation strategies, into maternal and child health clinics and pragmatically evaluate its clinical, service delivery, and implementation outcomes
Study Design	A 2-arm, non-blinded cluster randomized controlled trial (RCT) comparing the effect of IPMH and associated implementation strategies on depression, anxiety, quality of life, and adverse perinatal outcomes vs. control (standard of care) among Kenyan women
Study Population	Pregnant Kenyan women
Inclusion Criteria	-Receiving antenatal care at study facility ->=20 weeks gestation -Age>=14 -screen positive for PMAD symptoms (PHQ-2>=3 and/or GAD-2>=3)
Target Sample Size	2970, including 405 women living with HIV (WLWH)
Actual Enrollment	2383
Participants Completed Follow-up	960
Duration	Follow-up through 6 months postpartum
Date Study Open	September 19, 2023
Date of First Complete Enrollment	February 17, 2025
Estimated End of Study	July 31, 2028
Endpoints	Primary: Depression (Patient Health Questionnaire-9 [PHQ-9] score) and anxiety (Generalized Anxiety Disorder-7 [GAD-7] score) Secondary: -Problem Management Plus (PM+) Skill Acquisition (Reducing Tension Checklist score) -Quality of Life (World Health Organization Quality of Life [WHOQOL-BREF] score) -Any adverse pregnancy outcome

1.1 Study Timeline Since Initiation:

- May 12, 2023: Initial IRB approval
- Jun 05, 2023: Initial ERC approval
- Sept 19, 2023: RO1 Start date
- Oct 04, 2024: Research assistant (RA) training initiation
- Oct 11, 2024: 1st DSMB meeting
- Oct 23, 2024: Facility randomization
- Nov 06, 2024: Facility staff training initiation
- Feb 17, 2025: Perinatal women recruitment initiation
- May 19, 2025: 25% of planned enrollment achieved
- May 27, 2025: 25% of planned women living with HIV (WLWH) enrollment achieved
- August 1, 2025: Enrollment halted due to end of grant period
- Sept 22, 2025: Enrollment resumed after receiving NOA
- Nov 21, 2025: 2nd DSMB meeting
- Nov 24, 2025: 50% of planned enrollment achieved
- Dec 03, 2025: 50% of planned WLWH enrollment achieved
- Jan 8, 2026: Data collection temporarily halted due to REDCap server upgrade
- Jan 22, 2026: Data collection resumed after REDCap server upgrade
- Mar 30, 2026: 75% of planned WLWH enrollment achieved

1.2 Key Changes to Study Protocol

We requested for a modification from the Kenyatta National Hospital Ethical Research Committee on May 12, 2025 to include women from ≥ 20 weeks gestation because most of the participants enrolled from ≥ 28 weeks gestation were not completing the required 5 sessions of the Problem Management+ (PM+) psychotherapy that is offered as part of the intervention due to delivery. By modifying the inclusion criteria, our participants have adequate time to receive the required 5 sessions of PM+ and we will be able to evaluate the mechanism of action of PM+ which is a secondary outcome for the study.

2. Study Progress

2.1 Recruitment Timeline

We began recruiting pregnant women into the study on February 17, 2025. 2383 participants have been recruited into the study, meeting 80.2% of our enrollment target.

Of the enrolled participants, 1572 (66%) have completed their 6 weeks, 1290 (54.1%) have completed 14 weeks and 960 (40.3%) have completed 6 months postpartum follow-up visits.

In general, the enrollment had a good start and has progressed well despite challenges right before and into the enrollment like the stop work order, which led to HIV Implementing partners staff being away for one week (Feb 3, 2025 - Feb 7, 2025), just two weeks before the launch of study enrollment. Enrollment was halted for about three weeks (March 4, 2025 - March 26, 2025) at one site (Ober Kamoth sub-county hospital) because the lay providers who support screening clients for PMAD symptoms, a gateway into the study for potential study participants with PMAD symptoms, were away on leave. For two weeks (March 17, 2025 - March 28, 2025) at Gita Subcounty, the lay providers were not reporting to duty daily; however, this did not greatly impact enrollment at the site. Over this duration, we have been doing the enrollments despite the challenges with HIV Implementing Partner staff demotivation as a result of uncertainty with the continuation of their contracts.

We halted enrollment of new participants on August 1, 2025 after the grant period ended on July 31, 2025, pending approval of the Notice of Award from the National Institute of Health. We received the Notice of Award on September 8, 2025, and enrollments resumed on September 22, 2025. During this period, follow up visits at 6 weeks, 14 weeks, and 6 months postpartum continued.

In Siaya County, there was a nurse strike which began on September 18, 2025. The strike affected participant enrollment in 5 intervention and 4 control sites, where we had not enrolled new participants into the study since the start of the strike. However, in one control site in Siaya, the facility had hired a locum nurse for the maternal child health clinic, and enrollment was not affected. In all intervention sites, PM+ offered by lay providers and telepsychiatry offered by psychiatrists were not affected. The strike was officially called off on November 17, 2025. We resumed participant enrollment in the affected sites right after the strike.

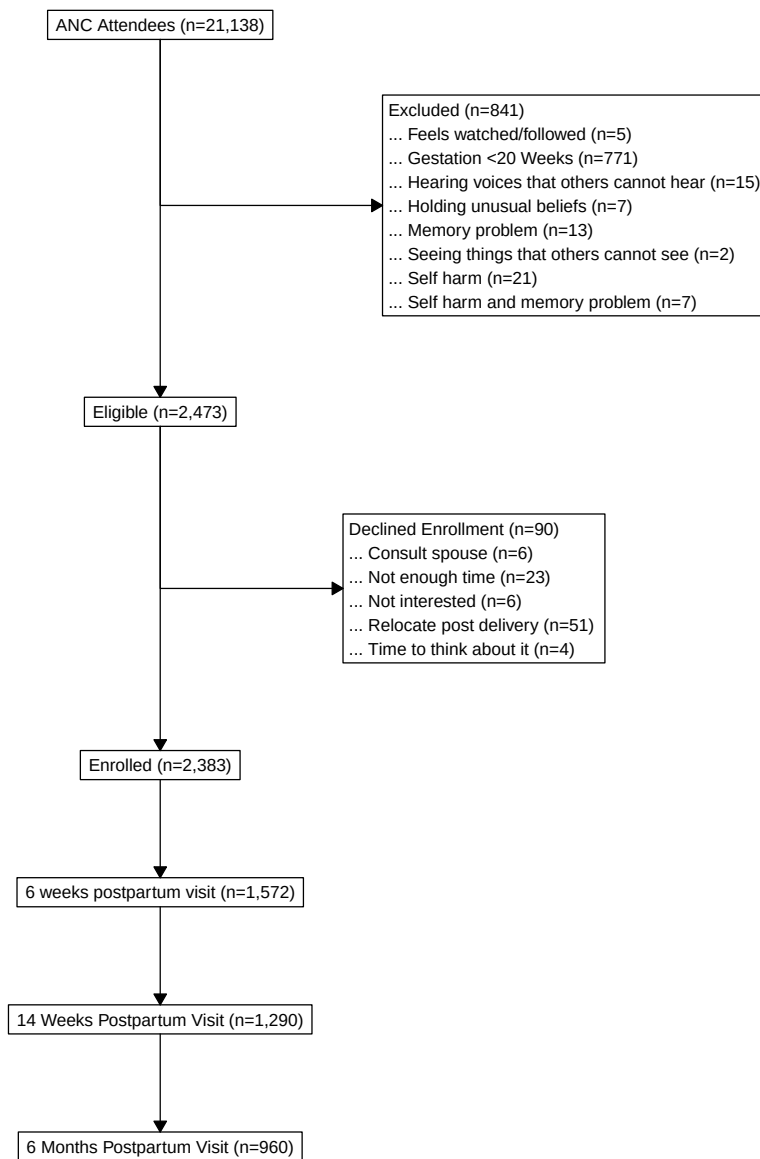
Data collection was also temporarily halted from Jan 8 to Jan 20, 2026 following the KNH REDCap server upgrade, to necessitate migration from the old system to the new one, as well as refreshing and testing of the data collection tool. During this period, we halted data collection. We resumed data collection on Jan 22, 2026.

3. Study Enrollment

3.1 Overall Cohort

3.1.1 Current subject status

So far, we screened 3314 women for eligibility among 21138 non-unique ANC attendees at recruitment sites. Of these, 2473 (74.6%) were eligible across intervention and control clusters. 2383 (96.4%) participants consented to participate in the study and were enrolled into the study. 1572 (66% out of all enrolled) completed their 6-weeks postpartum visit. 1290 (54.1% out of all enrolled) completed their 14-weeks postpartum visit, and 960 (40.3% out of all enrolled) completed their 6-months postpartum visit. Percentages do not reflect study retention as not all enrolled participants have reached follow-up visits

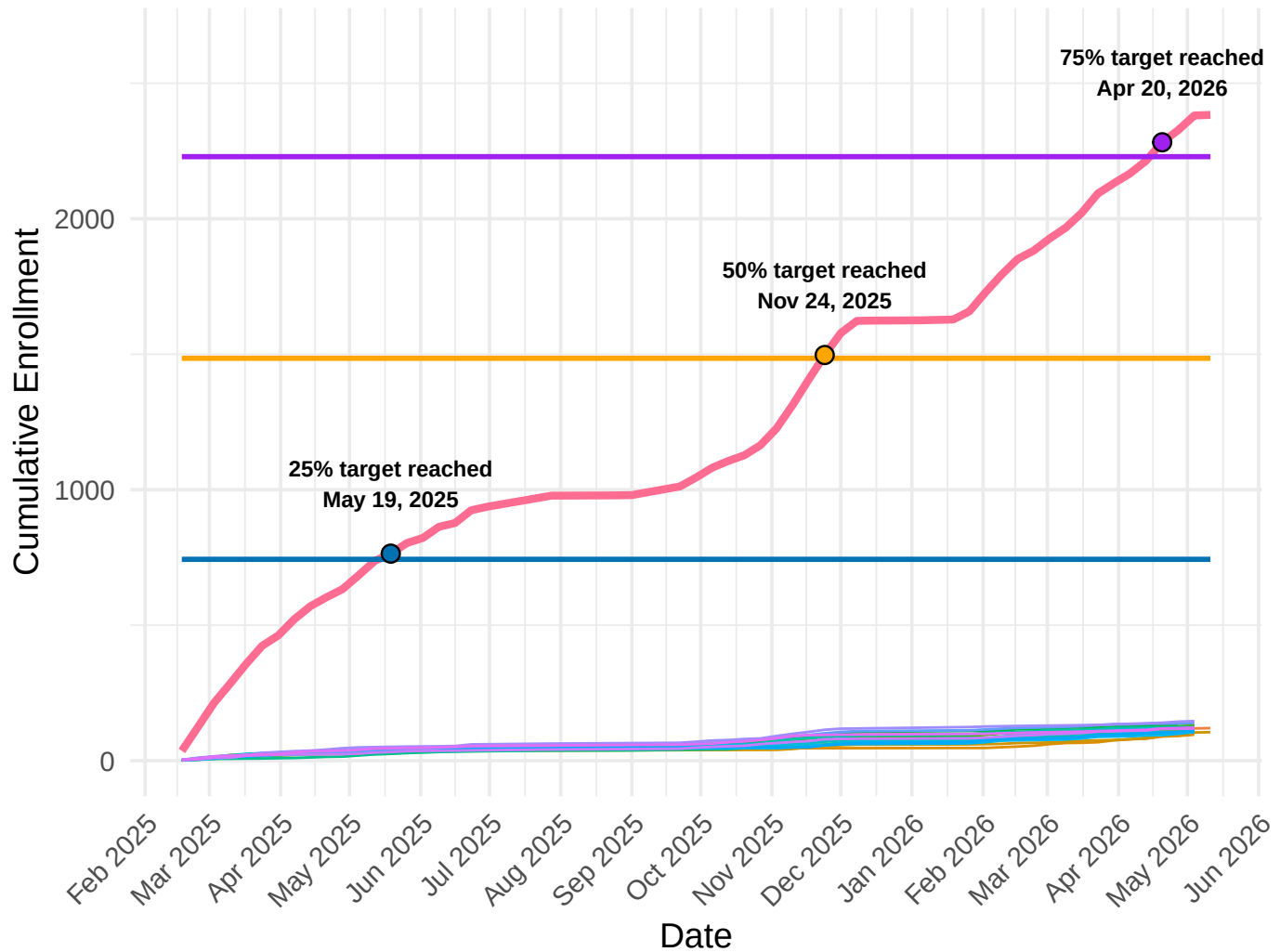


3.1.2 Enrollment Progress - Overall

On Feb 17, 2025 we began recruiting participants in all facilities. We hit our 25% enrollment target on May 19, 2025. We hit our 50% enrollment target on Nov 24, 2025. We hit our 75% enrollment target on Apr 20, 2026.

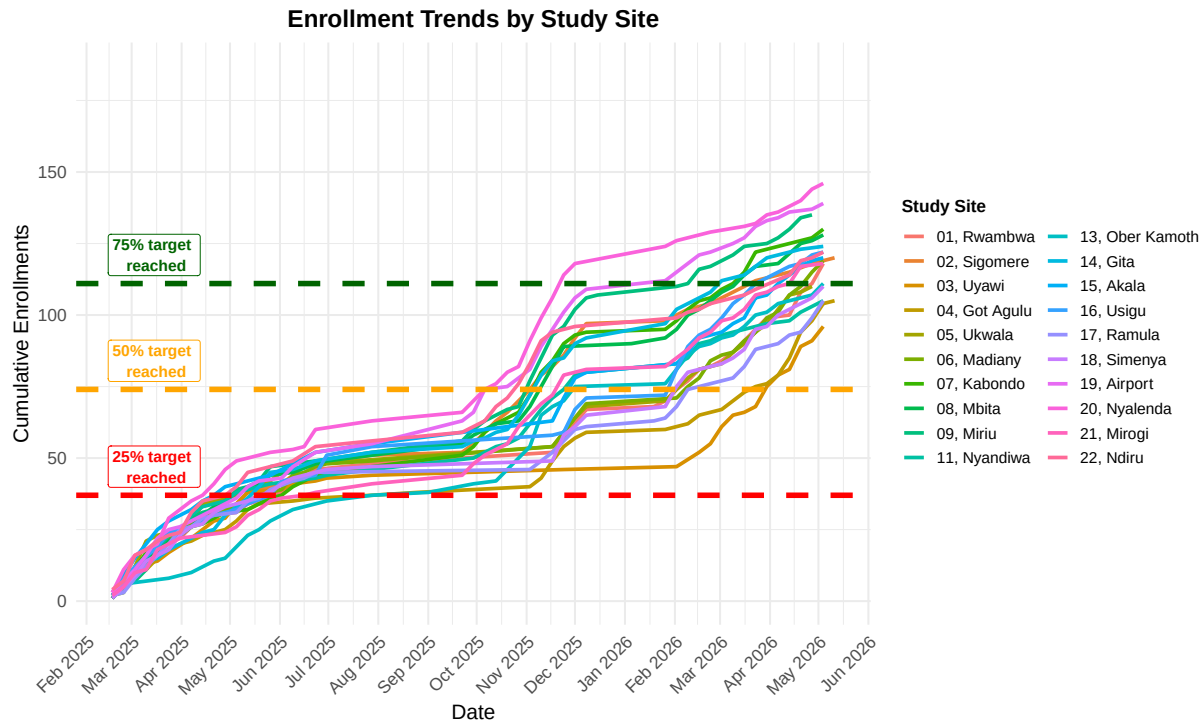
Cumulative Enrollment Over Time

Progress toward enrollment targets



3.1.3 Enrollment Progress - Per Facility

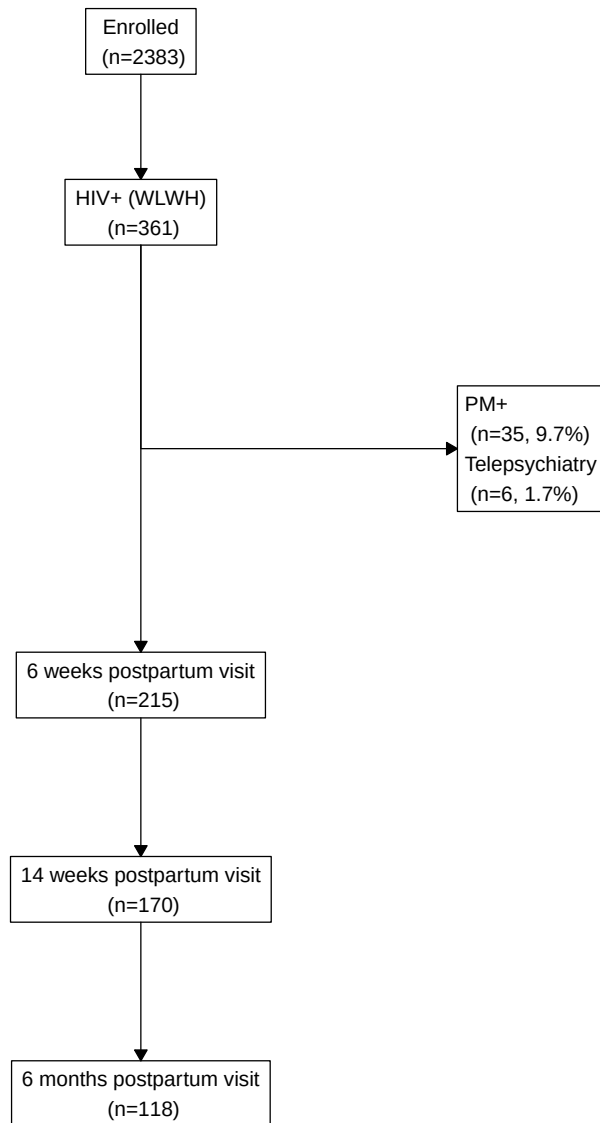
Enrollment progress across facility clusters has been steady and encouraging. As of May 8th 2025, 20 facility clusters have successfully reached the 75% enrollment milestone, corresponding to at least 111 participants per facility out of the targeted 148 perinatal women. The first facility to achieve this benchmark was Nyalenda Health Centre, underscoring its consistent leadership in enrollment performance. This achievement reflects strong commitment across clusters and positions the study well toward full enrollment.



3.2 Women Living with HIV

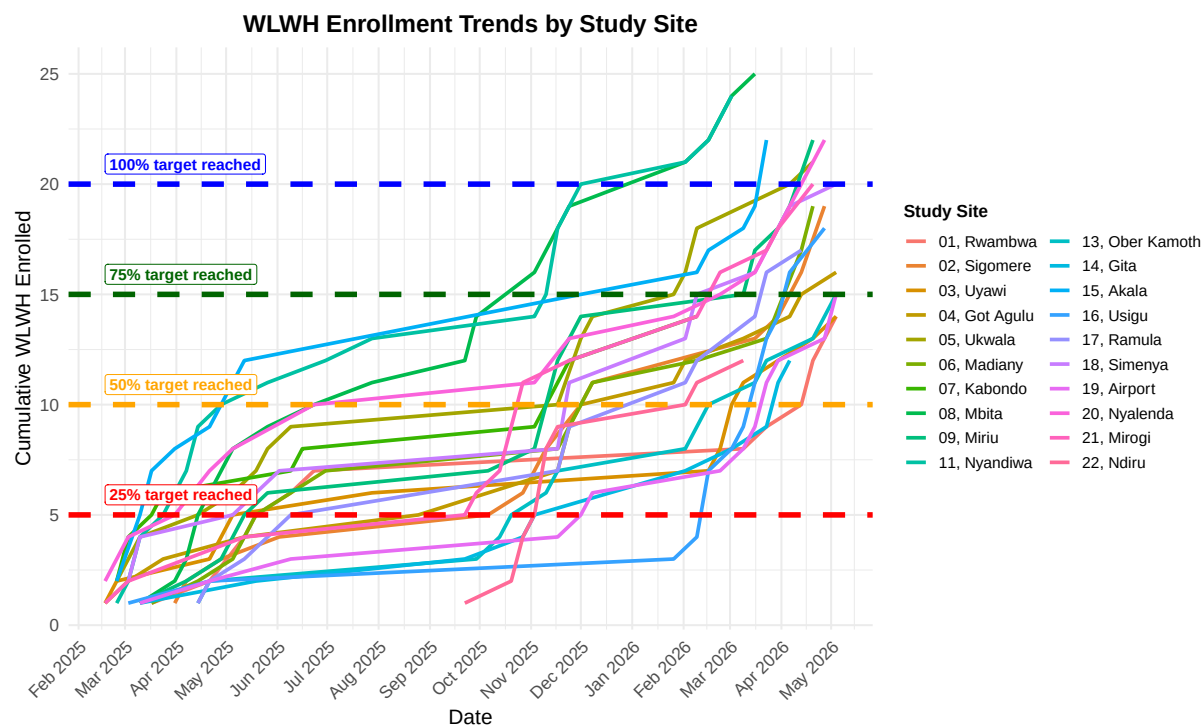
3.2.1 Current subject status

Overall 2383 participants have been enrolled into the study, of which 361 women are living with HIV (WLHIV), accounting to 15.1% of total enrollments. Of these, 215 (59.6%) have completed their 6 weeks, 170 (47.1%) have completed 14 weeks, and 118 (32.7%) have completed their 6 months postpartum follow-up visits. Percentages do not reflect study retention as not all enrolled participants have reached follow-up visits



3.2.2 Enrollment Progress

Recruitment of women living with HIV (WLWH) into the cohort continues to show steady improvement. We reached the 25% enrollment milestone for WLWH on May 27, 2025, the 50% enrollment milestone on Dec 03, 2025, and the 75% enrollment milestone on Mar 30, 2026. Out of the total enrollment target of approximately 20 WLWH per facility, 15 facilities have reached at least 75% of their target (15 participants), while 8 facilities have fully achieved their target of 20 participants. This progress demonstrates strong momentum across clusters and positions the study well toward complete enrollment.



4. Ineligibility

Women were ineligible if their gestational age at eligibility assessment was <20 weeks, if they endorsed thoughts/actions of self-harm, and if they endorsed symptoms of potential psychosis (e.g., hearing voices that others cannot hear, holding unusual beliefs, or feeling watched/followed), or if they endorsed experiencing memory problems.

Table 1: Reasons for Ineligibility

Characteristic	N = 841
Reasons for Ineligibility	
Gestation <20 Weeks	771 (91.7%)
Self harm	21 (2.5%)
Hearing voices that others cannot hear	15 (1.8%)
Memory problem	13 (1.5%)
Holding unusual beliefs	7 (0.8%)
Self harm and memory problem	7 (0.8%)
Feels watched/followed	5 (0.6%)
Seeing things that others cannot see	2 (0.2%)

¹ n (%)

5. Baseline Demographics

5.1 Participant Baseline Demographics

Table 2: Basic Demographic Summary

Characteristic	N	N = 2,383
Age (Years)	2,383	26.0 [IQR: 21.0, 31.0]
Do you currently have a partner	2,383	2,020 (84.8%)
Currently Married (Yes)	2,375	1,820 (76.6%)
Marriage (Monogamous)	1,820	1,601 (88.0%)
Shares residence with partner	2,020	1,664 (82.4%)
Currently in School	2,383	215 (9.0%)
Completed years in School	2,383	12.0 [IQR: 9.0, 14.0]
Do you have regular employment (Yes)	2,383	272 (11.4%)
Number of people in Household	2,383	4.0 [IQR: 3.0, 5.0]
Number of people sleep in same house	2,383	3.0 [IQR: 2.0, 5.0]
Rooms in the house most often sleep in	2,383	2.0 [IQR: 1.0, 3.0]
Time to the Clinic (Minutes)	2,383	30.0 [IQR: 20.0, 40.0]
Have you been pregnant before (Yes)	2,383	1,726 (72.4%)

¹ Median [IQR: Q1, Q3]; n (%)

* Eight participants preferred not to answer the marital status question and were excluded from N

5.2 WLWH Cohort Baseline Demographics

Table 3: WLWH Demographic Summary

Characteristic	N	N = 361
Age (Years)	361	30.0 [IQR: 25.0, 33.0]
Do you currently have a partner	361	313 (86.7%)
Currently Married (Yes)	357	290 (81.2%)
Marriage (Monogamous)	290	237 (81.7%)
Shares residence with partner (Yes)	313	264 (84.3%)
Currently in School	361	21 (5.8%)
Completed years in School	361	10.0 [IQR: 8.0, 12.0]
Do you have regular employment (Yes)	361	32 (8.9%)
Number of people in Household	361	5.0 [IQR: 3.0, 5.0]
Number of people sleep in same house	361	4.0 [IQR: 3.0, 5.0]
Rooms in the house most often sleep in	361	2.0 [IQR: 1.0, 3.0]
Time to the Clinic (Minutes)	361	30.0 [IQR: 20.0, 60.0]
Have you been pregnant before (Yes)	361	323 (89.5%)

¹ Median [IQR: Q1, Q3]; n (%)

* Four participants preferred not to answer the marital status question and were excluded from N

6. Study Terminations

To date, no participants have been withdrawn early due to discharge decisions by either the participant or the PI. A total of 960 participants have completed follow-up and exited the study at the scheduled endpoint. The first study exit occurred on Aug 18, 2025. Two maternal deaths (N=2) were reported; details are provided in the adverse events section.

7. Psychosocial Referrals

During enrollment and follow up study visits, participants are asked if they would like to be referred for psychosocial support if they screen positive for depression, anxiety and intimate partner violence (IPV). Participants at control sites are referred to routine care, while those at intervention sites are referred to telepsychiatry or PM+ based on clinical assessment of depression and/or anxiety. Out of 6280 total study visits, 982 (15.6%) participants have screened positive for referable conditions, of which, 897 (91.3%) accepted the referrals.

Table 4: Referral Uptake by Condition

Referral Condition	Screened +	Accepted	Uptake (%)
IPV Score $\geq$ 10	157	131	83.4
Anxiety Score $\geq$ 10	337	314	93.2
Depression Score $\geq$ 10	466	430	92.3
Self-harm	22	22	100.0

8. Primary and secondary study endpoints

The table below displays summaries of primary and secondary study endpoints for those who have completed their 14 week visit and/or who have experienced an adverse pregnancy outcome. Adverse pregnancy outcomes include miscarriage, stillbirth, preterm birth, low birth weight, small birth weight for gestational age, and infant/neonatal death. These include participants who may have not yet completed their 14 week visit.

Table 5: **Primary and Secondary Outcomes Summary**

Outcome	N	Outcome Statistics ¹
Patient Health Questionnaire-9 score ²	1,285	3.0 (1.0, 5.0)
Generalized Anxiety Disorder-7 score ²	1,288	3.0 (0.0, 5.0)
Quality of Life (WHOQOL BREF score) ²	1,289	75.0 (62.5, 75.0)
Any adverse pregnancy outcome	1,620	425 (26.2%)
Reducing Tensions Checklist score ²	1,288	17.0 (12.0, 22.0)

¹ Median (Q1, Q3); n (%)

² Participants with incomplete responses are excluded.

9. Adverse Events and Severe Adverse Events

9.1 Adverse Event Summary

As of 11 May 2026, 34 adverse events were reported. These will be included in the annual cumulative report to the ERC, in accordance with protocol requirements.

Table 6: Summary of Adverse Events

Characteristic	N = 34
Persistent or significant psychosocial distress	14 (41.2%)
Self-harm/Suicidal behavior	9 (26.5%)
Experienced violence or abuse	7 (20.6%)
Kicked out of home	3 (8.8%)
Injury	1 (2.9%)

* Persistent or significant psychosocial distress is defined as PHQ-9 score ≥ 10 at any follow-up visit.

9.2 Severe Adverse Event Summary

As of 11 May 2026, 54 serious adverse events were reported, of which 35 were infant deaths, 2 were maternal deaths, 15 were stillbirths, and 2 were a prolonged hospitalization.

Table 7: Summary of Reported Serious Adverse Events

Characteristic	N	N = 54
Maternal Death	54	2 (3.7%)
Maternal Death Cause	2	
Gastrointestinal bleeding		1 (50.0%)
Intraoperative complication during CS		1 (50.0%)
Infant Death	54	35 (64.8%)
Infants Death Cause	35	
Respiratory complications		18 (51.4%)
Birth asphyxia / Intrapartum hypoxia		4 (11.4%)
Fever/Vomiting		4 (11.4%)
Other / Unknown		3 (8.6%)
Prematurity		2 (5.7%)
Respiratory failure / Aspiration		2 (5.7%)
Congenital malformations / Birth defects		1 (2.9%)
Still birth		1 (2.9%)
Stillbirth	54	15 (27.8%)
New/prolonged hospitalization	54	2 (3.7%)

¹ n (%)